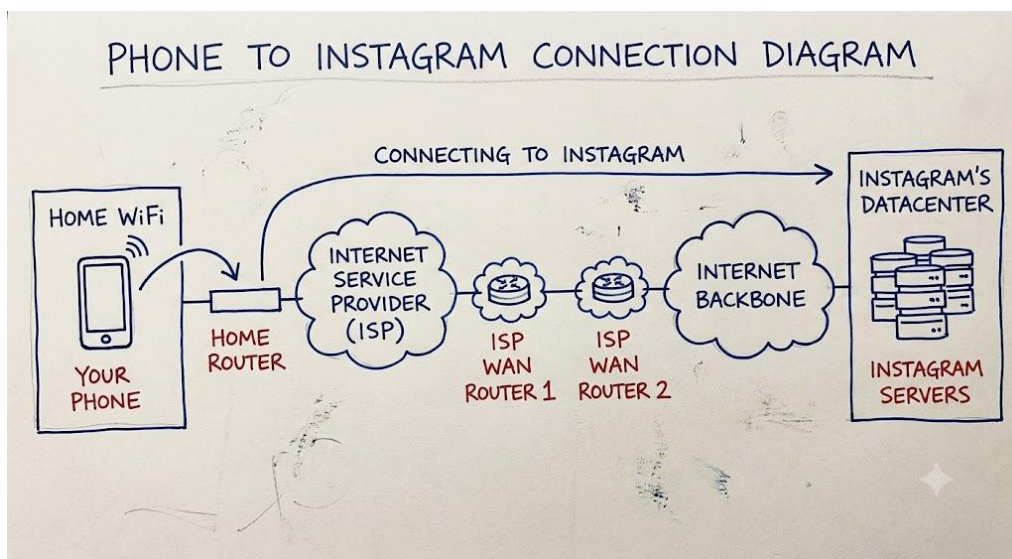


Wide Area Networks

Task 1 :- Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.

ANS :-



Task 2 :- Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

ANS :-

WAN Technology	Speed	Cost	Real-World Example
MPLS (Multi-Protocol Label Switching)	High & Guaranteed (10 Mbps to 10 Gbps)	High	IRCTC Ticketing : Connects regional branch offices and booking counters securely to central servers.
Leased Line	Extremely High & Dedicated (Up to 10 Gbps+)	Very High	Zomato Headquarters : Delivers fast, non-stop upload and download speeds for core business operations.
Broadband Internet	Variable / Shared (30 Mbps to 1 Gbps)	Very Low	Your Home : Provides simple internet access for everyday video streaming, social media, and online browsing.

Task 3 :- Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.

ANS :-

When connect to public coffee shop Wi-Fi, anyone on the same network can potentially snoop on your unencrypted traffic. A VPN (Virtual Private Network) protects your data using a WAN (Wide Area Network) :

How it works step-by-step :

1. Encrypted Tunnel Creation : When you turn on your VPN, it creates a secure, encrypted "tunnel" from your phone, through the coffee shop router, and across the ISP's WAN.
2. Hidden Traffic : Anyone spying on the public Wi-Fi or ISP network sees scrambled, unreadable code instead of your real activity or account credentials.
3. VPN Server Hand-off : Your encrypted data travels across WAN routers to a remote VPN Server. The VPN server decrypts your request and fetches your stream/data from Spotify or YouTube.
4. IP Masking & Return : Spotify/YouTube sends the video or music back to the VPN server, which encrypts it again and sends it securely back to your phone through the WAN tunnel.

[Your Phone] —(Encrypted Tunnel)—> [Public Wi-Fi / WAN] —> [VPN Server] —

Task 4 :- Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

ANS :-

Best Choice : SD-WAN (Software-Defined WAN) using standard broadband + 4G/5G backup.

2 Easy Reasons :

1. Auto-Backup : If one internet line goes down in Delhi, it instantly switches to a backup connection without stopping warehouse work.
2. Cheaper & Faster : It is much cheaper than private cables (MPLS) and super easy to add new warehouses as Flipkart grows.